

Predicting House Prices in the United States

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I - Introduction

House prices have long been an important and often controversial topic. While prices have been generally increasing since the 1970s (FRED), it can be very difficult to determine the price of a specific unit. There are a number of factors which can make such a prediction challenging, not all of them easily quantifiable. This project will seek to determine which variables are most significant in determining the price of a house, in the process developing an effective model of predicting house prices given several input variables.

Many studies have been conducted on the relationship between certain explanatory variables and house prices. From an economic standpoint, the primary factors are the supply and demand of houses at a given time. For instance, it is trivial that the number of houses available is significant in determining the price at which it is set to be sold. More specifically, it has been shown that factors such as location, number of bedrooms and bathrooms, mortgage interest rates, and other variables have a significant effect on the price of houses (Gerlach Real Estate). Perhaps the two most significant factors would be the current economic state of the US and the location of the house, but there is difficulty in adequately quantifying these variables, as they are certainly more qualitative in nature. The former could potentially be represented with indicator variables, while the latter could be represented by various metrics, such as GDP per capita, interest rate, unemployment rate, and others.

This project will explore the question of house pricing using several algorithms and computational techniques. Firstly, a linear regression analysis will be run to gain an initial understanding of the most significant variables affecting house price. Doing this will give an idea of which variables should be focused on during the application of algorithms. In terms of algorithms, it will apply the following: k-NN - a simple classification algorithm which predicts class according to the k closest data points, Naïve Bayes - which concerns conditional probabilities and involves parameter reduction, Decision Trees - which involve conditional statements whilst considering the chances of outcomes, and others as necessary. Each algorithm will be applied to the same dataset consisting of several input variables, and will then be assessed on its effectiveness. From that point, the best possible model will be selected and used to make predictions on new data points as well as to make projections for the future of house prices.

II - Motivation and Rationale

This project will certainly have a great deal of utility in that it will provide insight toward a major aspect of everyday life. Considering that housing and shelter are primary needs for all people, the information provided from this study has the potential to help readers from any demographic background. Not only will the project help to make predictions on unit prices given several input variables, but it will also make future projections, allowing readers to make informed decisions. The aforementioned controversy of the topic arises from the fact that house prices and thereby rent costs continue to increase while the wages of many continue to lag behind. Perhaps this project can provide some level of insight as to the reason for this trend, potentially helping readers to protect their wallets by making decisions based on thorough analyses.

An additional motivation for the project proposed is to gain experience in applying complex algorithmic methods to highly-applicable topics. This is to say that the project will likely help to develop important skills for upcoming statisticians / data analysts. Further, the problem is an interesting one to study because it has a plethora of potential answers. Several past studies and research documents have sought to adequately predict housing prices based on input variables, but often find difficulty due to outliers or to variables which are difficult to implement. Ideally, the project will help to resolve some of these issues by applying several complex classification algorithms.

III - Evaluation

Mathematically, the primary assessment of our project will be in the calculation of a success rate. That is, we will consider the mathematical effectiveness of the algorithms we implement within the project. Technically speaking, this will be done by utilizing a test set of data. The original data set will be split at random into two separate groups, one group (likely consisting of more than 70% of the data) will be used to train the model, and the other (likely 30% or less of the data) will be used to test the model. After testing the model, the number of correct predictions will be counted, ultimately giving a success rate. This rate should provide an idea of how effective the model was. In calculating this rate, the project will also seek to minimize the indicator loss function. A larger result from said function would suggest the model did not fit the data particularly well. Ideally the project will reduce overfitting and underfitting as much as possible. That is, the final model should be able to capture the underlying structure of the data and make reasonable, accurate, and consistent predictions on new data.

Non-mathematically, assessing the project might be more difficult as it would take time to see whether the projections were accurate. Generally speaking, a successful project is one that acknowledges its strengths as well as its weaknesses, and attempts to cover as many crucial aspects as possible. That is, the project will attempt to be extremely thorough, and part of its assessment will rely upon its level of thoroughness. The ideal project is one that is insightful as well as thorough, so upon completion there will be reflections upon its applications.

IV - Resources

- Dataset: House Prices with Advanced Regression Techniques (Kaggle)
-<https://www.kaggle.com/competitions/house-prices-regression-techniques>
- Programming Software: Google Colab (coding algorithms, displaying results)
-<https://colab.research.google.com/>
- Pandas, SciKit-Learn, and Matplotlib (to implement algorithms and plot figures)
- House Price and Variable Trends: Federal Reserve Economic Data (FRED)
-<https://fred.stlouisfed.org/series/ASPUS>
- Background Research: (additional resources)
-<https://www.brightmlshomes.com/the-major-variables-affecting-the-price-of-homes>
-<https://www.economicshelp.org/housing/factors-that-affect-the-housing-market/>
-<https://medium.com/@manilwagle/predicting-house-prices-using-machine-learning>

V - Contributions

- Alex: Organizing/writing documents, conducting background research, testing code/algorithms
- John: Data mining/cleaning, EDA, Model Selecting, and feature engineering
- Aaron: Conducting background research, analyzing algorithms, assessing project